A Forensic Analysis of a Memory Dump

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Note that Information contained in this document is for educational purposes.

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1 INTRODUCTION

1.1 BACKGROUND

An investigation was launched to forensically analyze a memory dump of a device's filesystem to ascertain information contained within it. Five questions were provided about the memory dump; the goal of this investigation is to answer each one using proper forensic methods and techniques without leaving any chances of contaminating the artefact. The five questions to answer are as follows:

1. What is the name of the logged in User?
2. What is the system time of the machine when the dump was taken?
3. In the "Technology Poem" open in Notepad, what colour of blue are the violets?
4. What Browser is visiting the webpage - "www.abertay.ac.uk"?
5. What webpage is the user visiting in the Microsoft Edge browser?

This investigation will be carried out on a windows machine using digital forensic tools and windows command line/ PowerShell.

1.2 AIM

The aims of this project exist to ensure that a thorough forensic investigation without allowing the memory dump to be modified in any way. The project aims to:

- Take proper forensic precautions to prevent the possibility of tampering with the memory dump.
- To carry out a full forensic investigation using trusted tools to answer all five of the questions provided.
- To document the entire investigation allowing reproducible results while showing that proper forensic methods were used so as not to invalidate the results.

2 PROCEDURE

2.1 OVERVIEW OF PROCEDURE

The investigative process started by taking a few precautions to ensure that the results remain valid and reliable. Initial information gathering was performed on the memory dump. Using this initial information gathering, the tester then took targeted approaches to each of the questions as this investigation does not require a full analysis of the memory dump. There was, however, some extra findings that were put in their own section later in the report as they were interesting and connected to the questions.

Digital forensic tools were used to complete the investigation and answer the provided questions. These tools were Volatility3 and MemProcFS. Volatility was used in the form of command line as well as the WorkBench GUI in some instances. The investigation started with using Volatility and moved onto MemProcFS if results were inconclusive or if further evidence was required.

2.2 PRECAUTIONS

In this investigation it is vital that the memory dump stays intact without being tempered with or modified in any way. Due to this, proper forensic protocol is used, taking multiple precautions before starting the investigation. The file was downloaded off OneDrive cloud storage, creating a local copy on the first machine. Seeing as parts of this investigation were carried out on different devices, a backup copy was also installed on the second machine, giving multiple options to gain another copy of the memory dump if either the cloud or second machine were unavailable for any reason. The following precautions were also carried out on both machines before any investigative actions were carried out on them to ensure that the forensics environment was secure regardless of the machine in use.

The first step was to take the hash of the memory dump file (*figures 1 & 11*). This way, the files hash can be checked at any point in the investigation to check if the file has been modified in any way. If the hash is the same at the end of the investigation, then it means that all the results were taken without any form of modification on memory dump. The hash was taken using SHA256 as this is a respected and more secure hashing algorithm than others like MD5.

```
C:\Users\Owner\Desktop\Volatility\Volatility\VolatilityWorkbench>CertUtil -hashfile Memory2203984Dump.dmp SHA256
SHA256 hash of Memory2203984Dump.dmp:
3bfa84343877e6c30a2b828aed5cd829e45db7ad234f5f7cf6c78df9f0fbac46
CertUtil: -hashfile command completed successfully.
```

Figure 1: PC 1 initial file hash

Next the file was set to read only as an extra layer of security against potential accidental modifications. This was done through PowerShell as can be seen (*figures 2 & 12*). This also means that other forensic tools can be used on the file without the fear of them modifying the file themselves.

```
C:\Users\Owner\Desktop\Volatility\Volatility\VolatilityWorkbench>powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Owner\Desktop\Volatility\Volatility\VolatilityWorkbench> Set-ItemProperty .\Memory2203984Dump.dmp -Name IsReadOnly -Value $true
PS C:\Users\Owner\Desktop\Volatility\Volatility\VolatilityWorkbench> Get-Item .\Memory2203984Dump.dmp


    Directory: C:\Users\Owner\Desktop\Volatility\Volatility\VolatilityWorkbench


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-ar---        19/10/2025     23:38     4272443392 Memory2203984Dump.dmp


PS C:\Users\Owner\Desktop\Volatility\Volatility\VolatilityWorkbench> Get-Item .\Memory2203984Dump.dmp | Select-Object Name, Attributes

Name                     Attributes
----                     ----------
Memory2203984Dump.dmp ReadOnly, Archive
```

Figure 2: PC 1 ReadOnly

2.3 Question 1

What is the name of the logged in User?

Volatility was used to run the pslist command which provides initial information gathering on what processes were running at the time the memory dump was taken (*figure 3 & Appendix B*).

```
C:\Users\Owner\Desktop\Volatility\Volatility\volatility3-develop>vol.py -f Memory2203984Dump.dmp windows.pslist
```

Figure 3: Volatility PsList

After that, the sessions command was run using Volatility WorkBench (*figure 4*). This showed the user "Xavier" was logged in as can be seen (*figure 13*).

```
"C:\Users\Owner\Desktop\Volatility\Volatility\VolatilityWorkbench\vol.exe" -f "C:\Users\Owner\Downloads\Memory2203984Dump.dmp" windows.sessions.Sessions
```

Figure 4: Volatility WorkBench Sessions

2.4 Question 2

What is the system time of the machine when the dump was taken?

The windows "info" command was run using Volatility WorkBench to answer this question (*figure 5*).

```
"C:\Users\Owner\Desktop\Volatility\Volatility\VolatilityWorkbench\vol.exe" -f "C:\Users\Owner\Downloads\Memory2203984Dump.dmp" windows.info.Info
```

Figure 5: Volatility WorkBench Info

The results show the system time to be 2nd of October 2025 at 20:18. This command also shows us some other information about the system such as its operating system being 64bit windows 10 (*figure 14*).

2.5 Question 3

In the "Technology Poem" open in Notepad, what colour of blue are the violets?

Initially Volatility was used to attempt to answer this question. The analyst used Volatility's YaraScan plugin to search the file system for certain string conditions. First, a Yara rule file was constructed to search for the keywords - *technology*, *poem* and *violet* as these seemed the most relevant to the little information known about this file at the time of testing. The file was saved as "poem.yar" (*figure 15*), making it a readable rule file for YaraScan to use.

This rule file was then used to search processes which could potentially be the notepad file in question. The process with PID 3408 was scanned (*figures 6 & 16*). This process showed many instances of containing the keywords "poem" and "violet" making it likely to at least be connected to notepad.

Figure 6: Volatility YaraScan 3408

The tester then swapped over to MemProcFS to investigate the file system directly for the text file. The drive was mounted in forensic mode (*figure 7*).

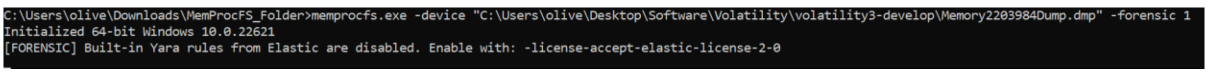


Figure 7: MemProcFS Mounting

After that a file explorer process was opened to navigate the mounted drive "M:\" (*figure 17*). From here, the "forensic\ntfs\1\" directory was entered (*figure 18*). MemProcFS attempts to recreate the memory dumps file system here, allowing the tester to explore the machine's files. From here a directory called "Poem" was entered which contained the file "Technology Poem.txt". After opening this file in notepad, the answer to the third question was found "Violets are Royal Blue - #9E1 (65, 105, 225)" (*figure 8*).

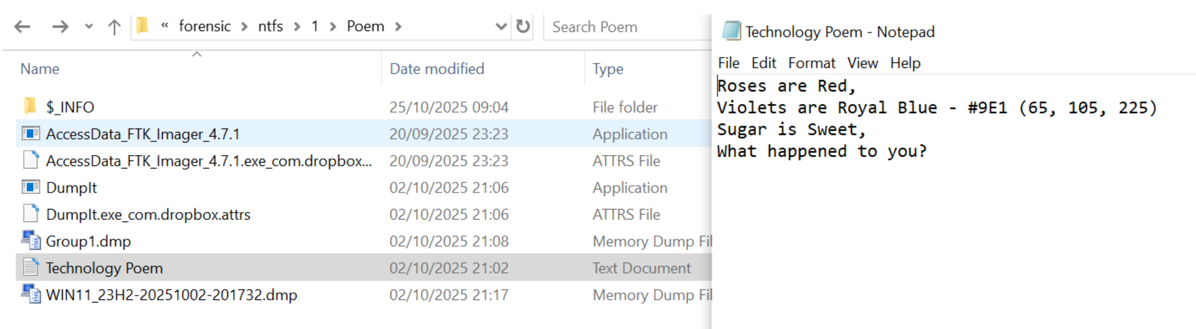


Figure 8: Technology Poem

2.6 Question 4

What Browser is visiting the webpage - "www.abertay.ac.uk"?

Using the MemProcFS "forensic\csv" directory, "process.csv" can be found (*figure 19*). This file is a spreadsheet of all the processes that were running at the time the memory dump was taken. This is used to gather information on what browsers were open on the machine. Browsers were searched for in the file using the excel search feature by typing in process names. The two browsers found were "msedge.exe" and "chrome.exe" with PID's 6512 and 7288 respectively (*figures 20 & 21*). Other browsers were searched for but not found include “firefox.exe”, “opera.exe”, “brave.exe” and “iexplore.exe”.

A dump was created for the two processes found using their PID’s with Volatility’s Memmap command (*figure 9*). Since Chrome seemed more suspicious to the tester, this was the dump that was analyzed initially. This was done by searching the files strings using Microsoft’s Sysinternal’s “strings.exe” (*figure 10*).

```
>python vol.py -f Memory2203984Dump.dmp windows.memmap --pid 7288 --dump
```

Figure 9: Volatility Memmap Chrome

```
C:\Users\Owner\Downloads\Strings>strings.exe C:\Users\Owner\Desktop\Volatility\Volatility\volatility3-develop\pid.7288.dmp | findstr /i "www.abertay.ac.uk"
```

Figure 10: Strings.exe Chrome

The tool found a substantial number of related strings to the website prompt “www.abertay.ac.uk” as seen (*figure 23*). This snippet contains a few crucial strings to answering this question, the tester will focus on this one:
“https://www.google.com/search?q=https+www+abertay+ac+uk&oq=https%3A%2F%2Fwww.aberta&sourceid=chrome&ie=UTF-8”.

This URL can be broken down to show that the site “https+www+abertay+ac+uk” (encoded version of “www.abertay.ac.uk”) was visited from the Google search engine. The most important section, however, is “sourceid=chrome” which is a flag added by Chrome when a search is performed using it.

2.7 Question 5

What webpage is the user visiting in the Microsoft Edge browser?

MemProcFS was once again used for this question. Since the drive was still mounted, the forensic file was further explored. This directory has a “timelines” directory holding text files containing various timelines of the memory dump’s contents in a raw informational format. However, by going back to the previous directory, the “csv” directory can be located which contains csv files of this same information and provides useful headers to help identify what the data is. One of the files contained within this directory was “timeline_web.csv” which contains a timeline of the processes visited in Microsoft Edge (“msedge.exe”) on the memory dump (*figure 22*). This shows that the user (Xavier) visited BrewDog’s website “https://www.brewdog.com/” on 2025-10-02 at 20:15:42. It also shows that the processes PID is 6512.

2.8 Further Investigation

This section is dedicated to the information ascertained from the investigation but not strictly relevant to the niche questions provided.

Xavier was found to be the logged in user at the time of the memory dump as discussed in **Question 1**. However, more information about the user and other users on the system were found using MemProcFS.

The tester went to the directory at “M:\sys\users\” where a text file was located called “users.txt”. This was opened in notepad to show Xavier and many other users on the system as well as their SIDs (*figures 24 & 25*).

The timelines directory in MemProcFS was further viewed, analyzing the “process_timeline.csv” file. This showed that the notepad file the tester was looking for had the PID 8776, and PPID 3408 (*figure 26*).

The investigation concluded here with hashes being taken of the memory dumps on both investigative PCs which both matched the initial file hash results (figures 27 & 28).

3 CONCLUSION

3.1 CONCLUSION

The investigation concluded having found a definitive answer to each of the provided questions on the memory dump. The memory dump's analysis was carried out using well respected tools with strong reputations ensuring forensic integrity. While either Volatility or MemProcFS could have been used individually to answer each of the five questions, using a combination allows another perspective of the memory dump when one tool is harvesting inconclusive results. There is also a factor of personal preference to the tester for which tool is used for what. Safety precautions were taken, following proper forensic practice, ensuring that no modification came to the memory dump or the results yielded from it. Even though the investigation was carried out on multiple devices, the hash function remaining the same at the start and end of the investigation on both devices proves that all results were harvested from the original unedited file.

APPENDICES

APPENDIX A - IMAGES

```
C:\Users\olive\Desktop\Software\Volatility\volatility3-develop>CertUtil -hashfile Memory2203984Dump.dmp  SHA256
SHA256 hash of Memory2203984Dump.dmp:
3bfa84343877e6c30a2b828aed5cd829e45db7ad234f5f7cf6c78df9f0fbac46
CertUtil: -hashfile command completed successfully.
```

Figure 11: PC 2 initial file hash

```
PS C:\Users\olive\Downloads> Set-ItemProperty .\Memory2203984Dump.dmp -Name IsReadOnly -Value $true
PS C:\Users\olive\Downloads> Get-Item .\Memory2203984Dump.dmp


    Directory: C:\Users\olive\Downloads


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-ar---        18/10/2025     14:01     4272443392 Memory2203984Dump.dmp


PS C:\Users\olive\Downloads> Get-Item .\Memory2203984Dump.dmp | Select-Object Name, Attributes

Name                      Attributes
----                      ----------
Memory2203984Dump.dmp ReadOnly, Archive
```

Figure 12: PC 2 Read Only

```
1       -       1332    dwm.exe /SYSTEM 2025-10-02 20:11:49.000000 UTC
1       -       5632    sihost.exe      Win11_23H2/Xavier       2025-10-02 20:13:03.000000 UTC
1       -       5664    svchost.exe     Win11_23H2/Xavier       2025-10-02 20:13:03.000000 UTC
1       -       5740    svchost.exe     Win11_23H2/Xavier       2025-10-02 20:13:03.000000 UTC
1       -       5784    svchost.exe     Win11_23H2/Xavier       2025-10-02 20:13:03.000000 UTC
1       -       6068    userinit.exe    -       2025-10-02 20:13:03.000000 UTC
1       -       6136    taskhostw.exe   Win11_23H2/Xavier       2025-10-02 20:13:04.000000 UTC
1       Console 3408    explorer.exe    Win11_23H2/Xavier       2025-10-02 20:13:04.000000 UTC
1       -       732     svchost.exe     Win11_23H2/Xavier       2025-10-02 20:13:15.000000 UTC
1       -       6016    svchost.exe     Win11_23H2/Xavier       2025-10-02 20:13:15.000000 UTC
1       -       5148    SearchHost.exe  -       2025-10-02 20:13:23.000000 UTC
1       -       3316    Widgets.exe     Win11_23H2/Xavier       2025-10-02 20:13:23.000000 UTC
1       -       4448    StartMenuExper  Win11_23H2/Xavier       2025-10-02 20:13:23.000000 UTC
1       -       5060    RuntimeBroker.  Win11_23H2/Xavier       2025-10-02 20:13:25.000000 UTC
1       -       6188    RuntimeBroker.  Win11_23H2/Xavier       2025-10-02 20:13:26.000000 UTC
1       -       6208    svchost.exe     Win11_23H2/Xavier       2025-10-02 20:13:26.000000 UTC
1       -       6744    dllhost.exe     Win11_23H2/Xavier       2025-10-02 20:13:29.000000 UTC
1       -       5400    ctfmon.exe      Win11_23H2/Xavier       2025-10-02 20:13:37.000000 UTC
1       -       7196    smartscreen.ex  Win11_23H2/Xavier       2025-10-02 20:13:39.000000 UTC
1       Console 7288    chrome.exe      Win11_23H2/Xavier       2025-10-02 20:13:39.000000 UTC
1       Console 7476    chrome.exe      Win11_23H2/Xavier       2025-10-02 20:13:41.000000 UTC
1       Console 7760    chrome.exe      Win11_23H2/Xavier       2025-10-02 20:13:45.000000 UTC
1       Console 7772    chrome.exe      Win11_23H2/Xavier       2025-10-02 20:13:45.000000 UTC
1       -       7844    chrome.exe      -       2025-10-02 20:13:46.000000 UTC
1       -       2620    taskhostw.exe   Win11_23H2/Xavier       2025-10-02 20:14:07.000000 UTC
1       -       5920    chrome.exe      -       2025-10-02 20:14:07.000000 UTC
1       Console 1272    SecurityHealth  Win11_23H2/Xavier       2025-10-02 20:14:07.000000 UTC
1       -       656     chrome.exe      -       2025-10-02 20:14:13.000000 UTC
1       -       6200    chrome.exe      -       2025-10-02 20:14:19.000000 UTC
1       -       7248    chrome.exe      -       2025-10-02 20:14:19.000000 UTC
1       -       1528    chrome.exe      -       2025-10-02 20:14:22.000000 UTC
1       -       6048    chrome.exe      -       2025-10-02 20:14:23.000000 UTC
1       Console 5040    chrome.exe      Win11_23H2/Xavier       2025-10-02 20:14:23.000000 UTC
1       Console 6512    msedge.exe      Win11_23H2/Xavier       2025-10-02 20:14:59.000000 UTC
1       Console 8048    msedge.exe      Win11_23H2/Xavier       2025-10-02 20:15:02.000000 UTC
1       Console 5724    msedge.exe      Win11_23H2/Xavier       2025-10-02 20:15:03.000000 UTC
1       Console 8008    msedge.exe      Win11_23H2/Xavier       2025-10-02 20:15:03.000000 UTC
1       -       6312    msedge.exe      -       2025-10-02 20:15:04.000000 UTC
N/A     -       1844    MemCompression  -       2025-10-02 20:11:49.000000 UTC
N/A     -       0               -       N/A
Time Stamp: Mon Oct 20 06:13:31 2025
```

Figure 13: Volatility Sessions

```
Variable        Value
Kernel Base     0xf80254c00000
DTB     0x1ae000
Symbols file:///C:/Users/Owner/Desktop/Volatility/Volatility/VolatilityWorkbench/symbols/windows/ntkrnlmp.pdb/621801E9DBEB95A617726F00B9A76710-1.json.xz
Is64Bit True
IsPAE   False
layer_name      0 WindowsIntel32e
memory_layer    1 WindowsCrashDump64Layer
base_layer      2 FileLayer
KdVersionBlock  0xf80255809998
Major/Minor     15.22621
MachineType     34404
KeNumberProcessors      4
SystemTime      2025-10-02 20:18:48+00:00
NtSystemRoot    C:\Windows
NtProductType   NtProductWinNt
NtMajorVersion  10
NtMinorVersion  0
PE MajorOperatingSystemVersion  10
PE MinorOperatingSystemVersion  0
PE Machine      34404
PE TimeDateStamp        Wed May 25 15:45:46 2033
Time Stamp: Mon Oct 20 06:52:39 2025
```

Figure 14: Volatility Info

```
rule poem_search {
        strings:
                $a = "technnology" nocase
                $b = "poem" nocase
                $c = "violet" nocase
        condition:
                $a or $b or $c
}
```

Figure 15: Yara Rule File "poem.yar"

```
50 6f 65 6d                                             Poem
0x188a7f7e      3408    poem_search     $b
50 6f 65 6d                                             Poem
0x188a7fde      3408    poem_search     $b
50 6f 65 6d                                             Poem
0x188acefe      3408    poem_search     $b
50 6f 65 6d                                             Poem
0x188ad5be      3408    poem_search     $b
50 6f 65 6d                                             Poem
0x188ad6de      3408    poem_search     $b
50 6f 65 6d                                             Poem
0x188ada9e      3408    poem_search     $b
50 6f 65 6d                                             Poem
0x188ade5e      3408    poem_search     $b
50 6f 65 6d                                             Poem
0x7fff134d094a  3408    poem_search     $c
56 69 6f 6c 65 74                                       Violet
0x7fff134d0ac1  3408    poem_search     $c
56 69 6f 6c 65 74                                       Violet
0x7fff134d0d8c  3408    poem_search     $c
56 69 6f 6c 65 74                                       Violet
0x7fff134d0e49  3408    poem_search     $c
56 69 6f 6c 65 74                                       Violet
0x7fff134d0f80  3408    poem_search     $c
76 69 6f 6c 65 74                                       violet
```

Figure 16: Yarascan 3408

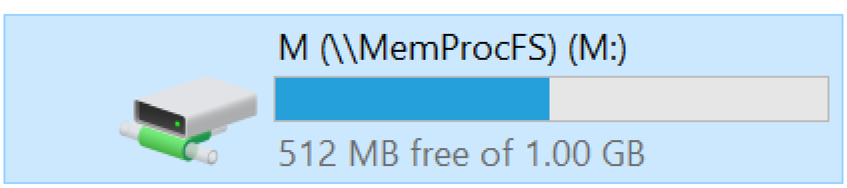


Figure 17: MemProcFS memory dump mounted

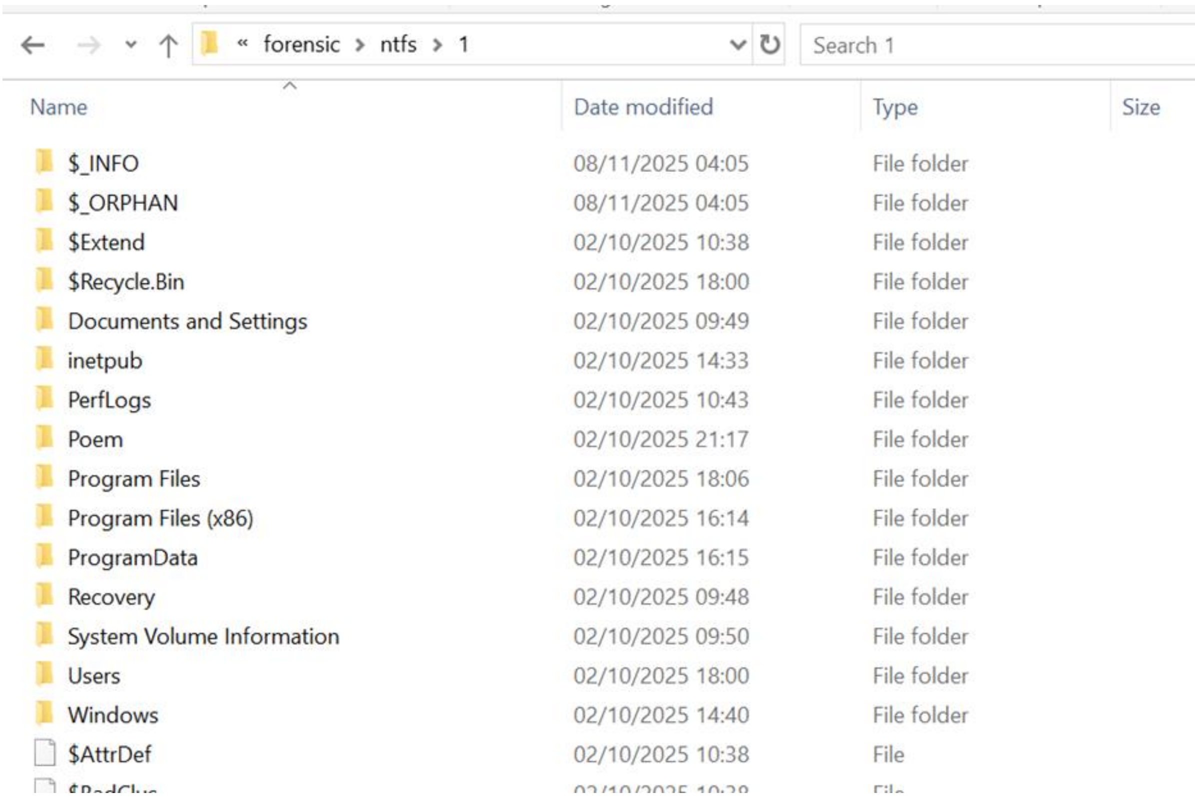


Figure 18: MemProcFS Reconstructed Filesystem

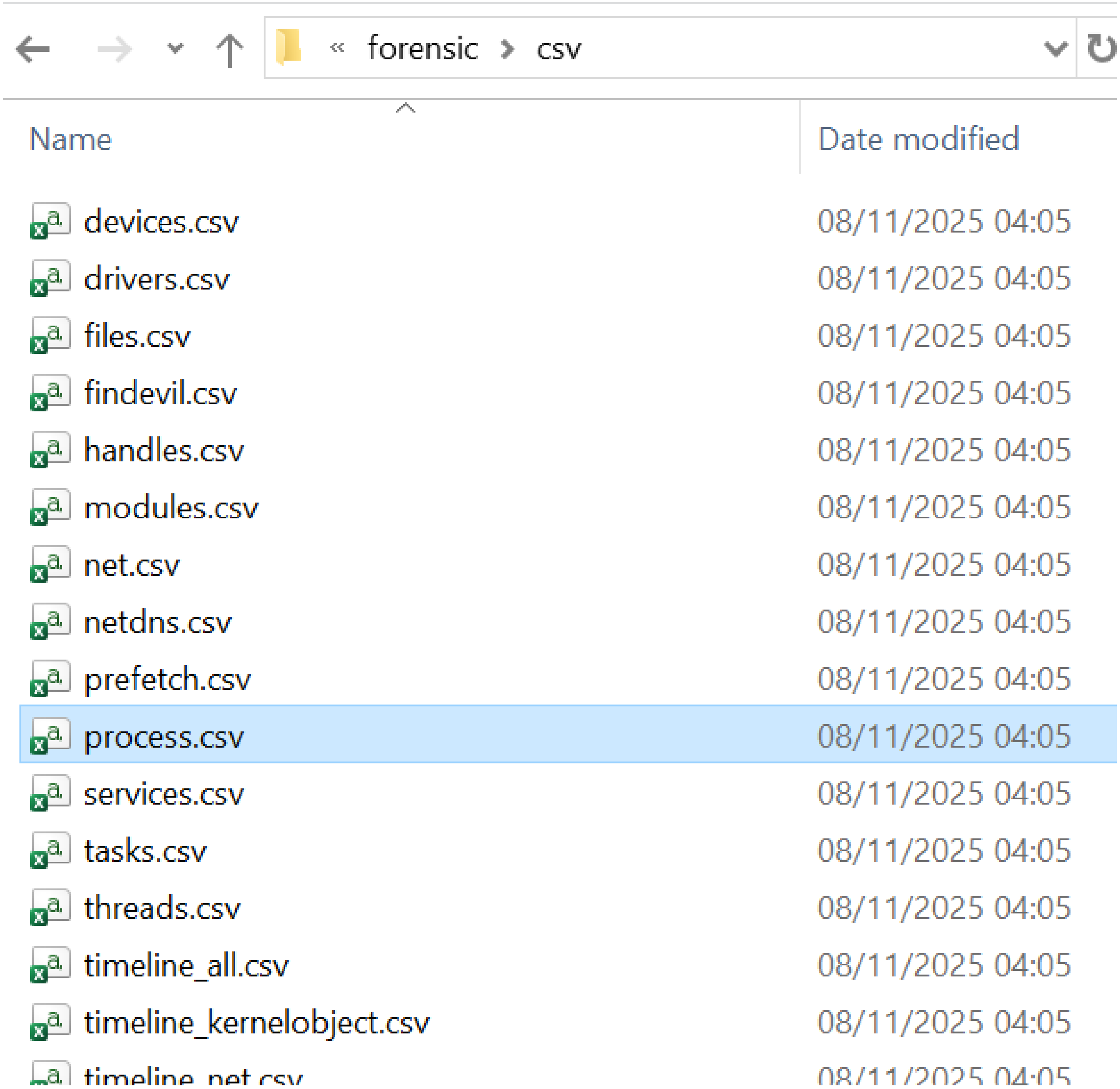


Figure 19: MemProcFS "csv" folder

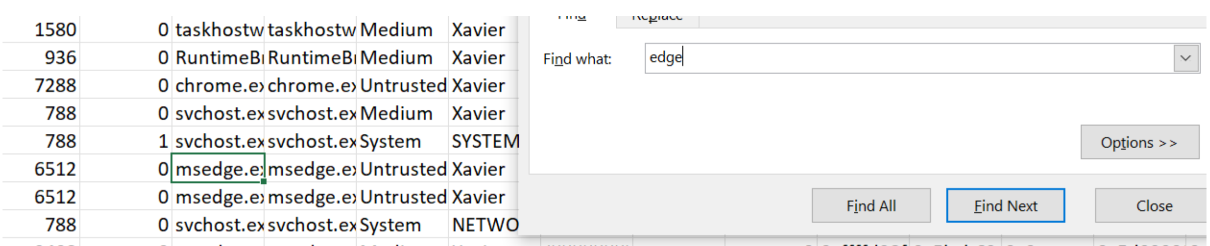
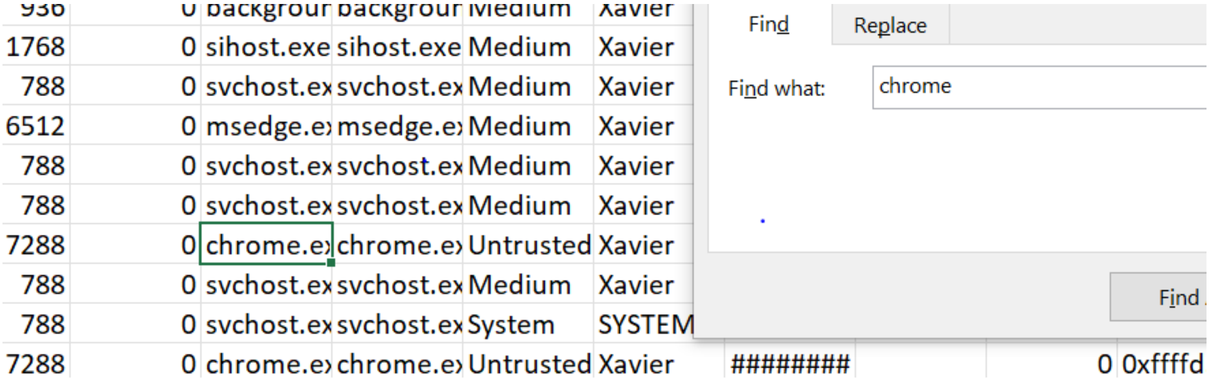


Figure 20: MemProcFS "process.csv" edge



936	0	backgrour	backgrour	Medium	Xavier					
1768	0	sihost.exe	sihost.exe	Medium	Xavier					
788	0	svchost.ex	svchost.ex	Medium	Xavier					
6512	0	msedge.e›	msedge.e›	Medium	Xavier					
788	0	svchost.ex	svchost.ex	Medium	Xavier					
788	0	svchost.ex	svchost.ex	Medium	Xavier					
7288	0	chrome.e›	chrome.e›	Untrusted	Xavier					
788	0	svchost.ex	svchost.ex	Medium	Xavier					
788	0	svchost.ex	svchost.ex	System	SYSTEM					
7288	0	chrome.e›	chrome.e›	Untrusted	Xavier		########		0	0xffffd

Figure 21: MemProcFS "process.csv" chrome

Time	Type	Action	PID	Value32	Value64	Text	P
########	WEB	CRE	6512	0x0	0x0	browser:[EDGE] type:[VISIT] url:[https://www.brewdog.com/] info:[BrewDog - Online Craft Beer Shop & Craft Beer Bars]	
########	WEB	CRE	6512	0x0	0x0	browser:[EDGE] type:[VISIT] url:[http://www.brewdog.com/] info:[BrewDog - Online Craft Beer Shop & Craft Beer Bars]	
########	WEB	CRE	6512	0x0	0x0	browser:[EDGE] type:[VISIT] url:[https://brewdog.com/] info:[BrewDog - Online Craft Beer Shop & Craft Beer Bars]	
########	WEB	CRE	6512	0x0	0x0	browser:[EDGE] type:[VISIT] url:[https://www.microsoft.com/en-gb/edge/welcome?form=MT00LJ&form=MT004A&OCID=MT004A&cs=155232	
########	WEB	CRE	6512	0x0	0x0	browser:[EDGE] type:[VISIT] url:[https://www.microsoft.com/edge/welcome?form=MT00LJ&form=MT004A&OCID=MT004A] info:[Welcome to	
########	WEB	CRE	6512	0x0	0x0	browser:[EDGE] type:[VISIT] url:[file:///C:/Windows/system32/oobe/FirstLogonAnim.html] info:[]	

Figure 22: MemProcFS "timeline_web.csv"

```
www.abertay.ac.uk
abertay.ac.uk
abertay.ac.uk
www.abertay.ac.uk
abertay.ac.uk
abertay.ac.uk
https://www.google.com/search?q=https+www+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=https+www+abertay+ac+uk&oq=https%3A%2F%2Fwww.aberta&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=https+www+abertay+ac+uk&oq=https%3A%2F%2Fwww.aberta&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=www+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac.uk&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=https+my+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac.u&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=https+my+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac.u&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=www+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac.u&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=www+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac.u&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=https+my+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac.&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=https+my+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac.&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=www+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac.&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=www+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac.&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=https+my+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=https+my+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=www+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=www+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=https+www+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.a&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=https+www+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.a&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=https+www+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.&sourceid=chrome&ie=UTF-8
https://www.googletagmanager.com/static/service_worker/5a20/sw_iframe.html?origin=https%3A%2F%2Fwww.abertay.ac.uk
https://www.google.com/search?q=https+my+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac.uk&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=https+my+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac.uk&sourceid=chrome&ie=UTF-8
https://www.google.com/search?q=www+abertay+ac+uk&oq=https%3A%2F%2Fwww.abertay.ac.uk&sourceid=chrome&ie=UTF-8
3A%2F%2Fwww.abertay.ac.uk&sourceid=chrome&ie=UTF-8
%2Fwww.abertay.ac.uk&sourceid=chrome&ie=UTF-8
3A%2F%2Fwww.abertay.ac.u&sourceid=chrome&ie=UTF-8
%2Fwww.abertay.ac.u&sourceid=chrome&ie=UTF-8
3A%2F%2Fwww.abertay.ac.&sourceid=chrome&ie=UTF-8
%2Fwww.abertay.ac.&sourceid=chrome&ie=UTF-8
3A%2F%2Fwww.abertay.ac&sourceid=chrome&ie=UTF-8
%2Fwww.abertay.ac&sourceid=chrome&ie=UTF-8
%3A%2F%2Fwww.abertay.a&sourceid=chrome&ie=UTF-8
%3A%2F%2Fwww.abertay.&sourceid=chrome&ie=UTF-8
www.abertay.ac.uk
https://www.abertay/
https://www.abertay.a/
https://www.abertay./
https://www.abertay.ac.uk/
www.abertay.ac.uk
abertay.ac.uk
www.abertay.ac.uk
https://www.abertay.ac.uk/
www.abertay.ac.uk
abertay.ac.uk
www.abertay.ac.uk
https://www.abertay.ac.uk/
```

Figure 23: Strings.exe 7288 Chrome dump

```
M:\sys\users>dir
 Volume in drive M is DOKAN
 Volume Serial Number is 1983-1116

 Directory of M:\sys\users

27/10/2025  13:08    <DIR>          .
27/10/2025  13:08    <DIR>          ..
25/10/2025  08:04             2,160 users.txt
               1 File(s)          2,160 bytes
               2 Dir(s)     536,870,912 bytes free

M:\sys\users>cat
'cat' is not recognized as an internal or external command,
operable program or batch file.

M:\sys\users>notepad users.txt
```

Figure 24: CMD Opening users.txt

users - Notepad

File Edit Format View Help

```
   # Username                             SID
-----------------------------------------
0000 Xavier                               S-1-5-21-2271638536-2473640809-1927150093-1003
0001 C:\Users\Alice                       S-1-5-21-2271638536-2473640809-1927150093-1013
0002 C:\Users\Samuel                      S-1-5-21-2271638536-2473640809-1927150093-1004
0003 C:\Users\Noah                        S-1-5-21-2271638536-2473640809-1927150093-1006
0004 C:\Users\Gabriel                     S-1-5-21-2271638536-2473640809-1927150093-1015
0005 C:\Users\Issac                       S-1-5-21-2271638536-2473640809-1927150093-1010
0006 C:\Users\Fiona                       S-1-5-21-2271638536-2473640809-1927150093-1012
0007 C:\Users\Benjamin                    S-1-5-21-2271638536-2473640809-1927150093-1005
0008 C:\Users\Zoe                         S-1-5-21-2271638536-2473640809-1927150093-1008
0009 C:\Users\Daniel                      S-1-5-21-2271638536-2473640809-1927150093-1016
000a C:\Users\Yara                        S-1-5-21-2271638536-2473640809-1927150093-1007
000b C:\Users\Victor                      S-1-5-21-2271638536-2473640809-1927150093-1009
000c C:\Users\Emma                        S-1-5-21-2271638536-2473640809-1927150093-1002
000d C:\Users\Chloe                       S-1-5-21-2271638536-2473640809-1927150093-1014
000e C:\Users\Thomas                      S-1-5-21-2271638536-2473640809-1927150093-1011
000f C:\Users\Admin                       S-1-5-21-2271638536-2473640809-1927150093-1001
```

Figure 25: Notepad users.txt

144	8472	6512	0	msedge.e›	msedge.e›	Untrusted	Xavier	########	0	0xffffd38fa	0x7cf0c7e	0x0	0x1261b2(	0x136bb1(	C:\Prograr	\Device\H:	"C:\Progra	E
145	8776	3408	0	Notepad.e	Notepad.e	Medium	Xavier	########	0	0xffffd38fa	0x89e3c97	0x0	0x10016f0	0x3456e0(	C:\Prograr	\Device\H:	"C:\Progra	E
146	8780	6512	0	msedge.e›	msedge.e›	Low	Xavier	########	0	0xffffd38fa	0xd6f85ae	0x0	0x6df1e00	0x12731d(	C:\Prograr	\Device\H:	"C:\Progra	E

Figure 26: Notepad.exe PPID

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Owner\Desktop\Volatility\Volatility\volatility3-develop>CertUtil -hashfile Memory2203984Dump.dmp SHA256
SHA256 hash of Memory2203984Dump.dmp:
3bfa84343877e6c30a2b828aed5cd829e45db7ad234f5f7cf6c78df9f0fbac46
CertUtil: -hashfile command completed successfully.

C:\Users\Owner\Desktop\Volatility\Volatility\volatility3-develop>
```

Figure 27: PC1 Memory Dump Hash Final

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\olive\Desktop\Software\Volatility\volatility3-develop>CertUtil -hashfile Memory2203984Dump.dmp SHA256
SHA256 hash of Memory2203984Dump.dmp:
3bfa84343877e6c30a2b828aed5cd829e45db7ad234f5f7cf6c78df9f0fbac46
CertUtil: -hashfile command completed successfully.

C:\Users\olive\Desktop\Software\Volatility\volatility3-develop>
```

Figure 28: PC2 Memory Dump Hash Final

APPENDIX B – PSLIST RESULTS

Volatility 3 Framework 2.27.0

PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime File output

4 0 System 0xd38fa56ee040 178 - N/A False 2025-10-02 20:11:32.000000 UTC N/A Disabled
104 4 Registry 0xd38fa5798040 4 - N/A False 2025-10-02 20:11:29.000000 UTC N/A Disabled
440 4 smss.exe 0xd38fa8b3c040 2 - N/A False 2025-10-02 20:11:32.000000 UTC N/A Disabled
568 556 csrss.exe 0xd38fa9810140 12 - 0 False 2025-10-02 20:11:48.000000 UTC N/A Disabled

640	556	wininit.exe	0xd38fa9952080	6	-	0	False	2025-10-02 20:11:48.000000 UTC	N/A	Disabled
660	632	csrss.exe	0xd38fa9958080	16	-	1	False	2025-10-02 20:11:48.000000 UTC	N/A	Disabled
716	632	winlogon.exe	0xd38fa997f080	9	-	1	False	2025-10-02 20:11:48.000000 UTC	N/A	Disabled
788	640	services.exe	0xd38fa99ab080	10	-	0	False	2025-10-02 20:11:48.000000 UTC	N/A	Disabled
812	640	lsass.exe	0xd38fa99bf080	18	-	0	False	2025-10-02 20:11:48.000000 UTC	N/A	Disabled
936	788	svchost.exe	0xd38fa9a640c0	26	-	0	False	2025-10-02 20:11:48.000000 UTC	N/A	Disabled
964	716	fontdrvhost.ex	0xd38fa9a6e140	6	-	1	False	2025-10-02 20:11:48.000000 UTC	N/A	Disabled
972	640	fontdrvhost.ex	0xd38fa9a6c140	6	-	0	False	2025-10-02 20:11:48.000000 UTC	N/A	Disabled
524	788	svchost.exe	0xd38fa9ae70c0	11	-	0	False	2025-10-02 20:11:48.000000 UTC	N/A	Disabled
876	788	svchost.exe	0xd38fa9b1e080	6	-	0	False	2025-10-02 20:11:48.000000 UTC	N/A	Disabled
1036	788	svchost.exe	0xd38fa9b5a080	12	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
1084	788	svchost.exe	0xd38fa9ba5080	4	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
1092	788	svchost.exe	0xd38fa9bac0c0	3	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
1100	788	svchost.exe	0xd38fa9baf080	5	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
1204	788	svchost.exe	0xd38fa9c020c0	5	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
1228	788	svchost.exe	0xd38fa9c0a080	15	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
1332	716	dwm.exe	0xd38fa9c90080	29	-	1	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
1376	788	svchost.exe	0xd38fa9ca00c0	5	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
1388	788	svchost.exe	0xd38fa9c9a080	7	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
1428	788	svchost.exe	0xd38fa9cb10c0	4	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
1444	788	svchost.exe	0xd38fa9cb9080	6	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
1468	788	svchost.exe	0xd38fa9cbf080	5	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled

1492 788 svchost.exe 0xd38fa9cdc080 4 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
1580 788 svchost.exe 0xd38fa9d30080 16 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
1628 788 svchost.exe 0xd38fa9d66080 5 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
1688 788 svchost.exe 0xd38fa9d980c0 2 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
1768 788 svchost.exe 0xd38fa9db8080 7 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
1784 788 svchost.exe 0xd38fa9db6080 4 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
1804 788 svchost.exe 0xd38fa9dc3080 8 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
1844 4 MemCompression 0xd38fa9dcc040 72 - N/A False
2025-10-02 20:11:49.000000 UTC N/A Disabled
1852 788 svchost.exe 0xd38fa9dc9080 3 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
1972 788 svchost.exe 0xd38fa9e75080 11 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
1052 788 svchost.exe 0xd38fa9ed5080 6 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
1908 788 svchost.exe 0xd38fa9f17080 5 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
2136 788 svchost.exe 0xd38fa9f4c080 5 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
2152 788 svchost.exe 0xd38fa9f50080 4 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
2268 788 svchost.exe 0xd38fa9fed080 5 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
2288 788 svchost.exe 0xd38fa9ff20805 - 0 False 2025-10-02
20:11:49.000000 UTC N/A Disabled
2368 788 svchost.exe 0xd38faa020080 5 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
2484 788 svchost.exe 0xd38faa0b5080 4 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
2684 788 spoolsv.exe 0xd38faa1510c0 9 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
2732 788 svchost.exe 0xd38faa1a1080 20 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
2776 788 svchost.exe 0xd38faa2020c0 6 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled
2792 788 svchost.exe 0xd38faa206080 9 - 0 False
2025-10-02 20:11:49.000000 UTC N/A Disabled

2896	788	svchost.exe	0xd38faa242080	2	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
2904	788	svchost.exe	0xd38faa2570c0	15	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
2912	788	svchost.exe	0xd38faa25c080	15	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
2952	788	svchost.exe	0xd38faa261080	5	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
3016	788	svchost.exe	0xd38faa2c0080	3	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
3036	788	svchost.exe	0xd38faa2ca080	17	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
3048	788	svchost.exe	0xd38faa2c9080	6	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
1732	788	svchost.exe	0xd38faa2f1080	7	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
3180	788	blnsvr.exe	0xd38faa3ad080	5	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
3260	788	MsMpEng.exe	0xd38faa3f7080	37	-	0	False	2025-10-02 20:11:49.000000 UTC	N/A	Disabled
3568	2896	dasHost.exe	0xd38faa5230c0	2	-	0	False	2025-10-02 20:11:50.000000 UTC	N/A	Disabled
3656	788	svchost.exe	0xd38faa5df080	3	-	0	False	2025-10-02 20:11:51.000000 UTC	N/A	Disabled
3660	788	svchost.exe	0xd38faa584080	2	-	0	False	2025-10-02 20:11:51.000000 UTC	N/A	Disabled
3800	788	svchost.exe	0xd38faa811080	6	-	0	False	2025-10-02 20:11:51.000000 UTC	N/A	Disabled
3820	788	svchost.exe	0xd38faa678080	6	-	0	False	2025-10-02 20:11:51.000000 UTC	N/A	Disabled
4044	788	svchost.exe	0xd38faa67b080	6	-	0	False	2025-10-02 20:11:53.000000 UTC	N/A	Disabled
3124	788	svchost.exe	0xd38faa89e080	3	-	0	False	2025-10-02 20:11:53.000000 UTC	N/A	Disabled
3336	788	svchost.exe	0xd38faa6ca080	14	-	0	False	2025-10-02 20:11:53.000000 UTC	N/A	Disabled
2696	788	svchost.exe	0xd38faa67f080	37	-	0	False	2025-10-02 20:11:54.000000 UTC	N/A	Disabled
4168	2904	AggregatorHost	0xd38faa993080	4	-	0	False	2025-10-02 20:11:54.000000 UTC	N/A	Disabled
4272	2896	dasHost.exe	0xd38faa7c70c0	5	-	0	False	2025-10-02 20:11:55.000000 UTC	N/A	Disabled
4344	788	svchost.exe	0xd38faa7b7080	9	-	0	False	2025-10-02 20:11:56.000000 UTC	N/A	Disabled

4400	788	MpDefenderCore	0xd38faaad0080	9	-	0	False	2025-10-02 20:11:56.000000 UTC	N/A	Disabled
4012	936	WmiPrvSE.exe	0xd38faab16080	11	-	0	False	2025-10-02 20:11:58.000000 UTC	N/A	Disabled
4772	788	svchost.exe	0xd38faab15080	7	-	0	False	2025-10-02 20:11:59.000000 UTC	N/A	Disabled
2940	788	NisSrv.exe	0xd38faadd5080	12	-	0	False	2025-10-02 20:12:02.000000 UTC	N/A	Disabled
4224	788	svchost.exe	0xd38faade5080	22	-	0	False	2025-10-02 20:12:03.000000 UTC	N/A	Disabled
3432	936	WmiPrvSE.exe	0xd38faa464080	7	-	0	False	2025-10-02 20:12:09.000000 UTC	N/A	Disabled
5492	788	svchost.exe	0xd38faac7d0c0	12	-	0	False	2025-10-02 20:13:02.000000 UTC	N/A	Disabled
5632	1768	sihost.exe	0xd38faa445080	17	-	1	False	2025-10-02 20:13:03.000000 UTC	N/A	Disabled
5664	788	svchost.exe	0xd38fa8e6e0c0	6	-	1	False	2025-10-02 20:13:03.000000 UTC	N/A	Disabled
5740	788	svchost.exe	0xd38fa94310c0	5	-	1	False	2025-10-02 20:13:03.000000 UTC	N/A	Disabled
5784	788	svchost.exe	0xd38fa94330c0	13	-	1	False	2025-10-02 20:13:03.000000 UTC	N/A	Disabled
6068	716	userinit.exe	0xd38faa44e080	0	-	1	False	2025-10-02 20:13:03.000000 UTC	2025-10-02 20:13:19.000000 UTC	Disabled
6136	1580	taskhostw.exe	0xd38fa8999080	9	-	1	False	2025-10-02 20:13:04.000000 UTC	N/A	Disabled
3552	788	svchost.exe	0xd38fa8c4d080	9	-	0	False	2025-10-02 20:13:04.000000 UTC	N/A	Disabled
3408	6068	explorer.exe	0xd38fa8ed7080	98	-	1	False	2025-10-02 20:13:04.000000 UTC	N/A	Disabled
3424	788	svchost.exe	0xd38fa895e0c0	5	-	0	False	2025-10-02 20:13:05.000000 UTC	N/A	Disabled
1384	788	svchost.exe	0xd38fab1e00c0	9	-	0	False	2025-10-02 20:13:06.000000 UTC	N/A	Disabled
2176	788	svchost.exe	0xd38fa97040c0	5	-	0	False	2025-10-02 20:13:12.000000 UTC	N/A	Disabled
4412	788	svchost.exe	0xd38faae42080	13	-	0	False	2025-10-02 20:13:12.000000 UTC	N/A	Disabled
6024	788	svchost.exe	0xd38faa0ee080	5	-	0	False	2025-10-02 20:13:15.000000 UTC	N/A	Disabled
4440	788	SearchIndexer.	0xd38fa945c080	11	-	0	False	2025-10-02 20:13:15.000000 UTC	N/A	Disabled
732	788	svchost.exe	0xd38fa8f24080	6	-	1	False	2025-10-02 20:13:15.000000 UTC	N/A	Disabled

6016 788 svchost.exe 0xd38fa945f080 7 - 1 False
2025-10-02 20:13:15.000000 UTC N/A Disabled
2880 788 svchost.exe 0xd38fa93200c0 1 - 0 False
2025-10-02 20:13:21.000000 UTC N/A Disabled
5148 936 SearchHost.exe 0xd38faafe2080 66 - 1 False
2025-10-02 20:13:23.000000 UTC N/A Disabled
3316 936 Widgets.exe 0xd38fab3850c0 24 - 1 False
2025-10-02 20:13:23.000000 UTC N/A Disabled
4448 936 StartMenuExper 0xd38fab2f5080 23 - 1 False
2025-10-02 20:13:23.000000 UTC N/A Disabled
5060 936 RuntimeBroker.0xd38fab5bd0c0 9 - 1 False
2025-10-02 20:13:25.000000 UTC N/A Disabled
6188 936 RuntimeBroker.0xd38fab66c0c0 8 - 1 False
2025-10-02 20:13:26.000000 UTC N/A Disabled
6208 788 svchost.exe 0xd38fab7480c0 4 - 1 False
2025-10-02 20:13:26.000000 UTC N/A Disabled
6224 788 svchost.exe 0xd38fab5bf0c0 0 - 0 False
2025-10-02 20:13:26.000000 UTC 2025-10-02 20:17:39.000000 UTC Disabled
6744 936 dllhost.exe 0xd38fab93f080 10 - 1 False
2025-10-02 20:13:29.000000 UTC N/A Disabled
5400 1908 ctfmon.exe 0xd38fab4cc080 25 - 1 False
2025-10-02 20:13:37.000000 UTC N/A Disabled
7196 936 smartscreen.ex 0xd38fab4d2080 7 - 1 False
2025-10-02 20:13:39.000000 UTC N/A Disabled
7288 3408 chrome.exe 0xd38fac4850c0 44 - 1 False
2025-10-02 20:13:39.000000 UTC N/A Disabled
7476 7288 chrome.exe 0xd38fa8e130c0 8 - 1 False
2025-10-02 20:13:41.000000 UTC N/A Disabled
7612 788 svchost.exe 0xd38fab962100 5 - 0 False
2025-10-02 20:13:44.000000 UTC N/A Disabled
7760 7288 chrome.exe 0xd38fab87c080 11 - 1 False
2025-10-02 20:13:45.000000 UTC N/A Disabled
7772 7288 chrome.exe 0xd38fab879080 18 - 1 False
2025-10-02 20:13:45.000000 UTC N/A Disabled
7844 7288 chrome.exe 0xd38fac48c080 9 - 1 False
2025-10-02 20:13:46.000000 UTC N/A Disabled
7264 788 svchost.exe 0xd38fa9fea080 1 - 0 False
2025-10-02 20:13:56.000000 UTC N/A Disabled
6412 788 svchost.exe 0xd38fac5c5080 10 - 0 False
2025-10-02 20:14:02.000000 UTC N/A Disabled
2620 1580 taskhostw.exe 0xd38fab1de0c0 3 - 1 False
2025-10-02 20:14:07.000000 UTC N/A Disabled
3776 788 svchost.exe 0xd38fa898c080 7 - 0 False
2025-10-02 20:14:07.000000 UTC N/A Disabled

5920 7288 chrome.exe 0xd38fabc020c0 8 - 1 False
2025-10-02 20:14:07.000000 UTC N/A Disabled
1272 3408 SecurityHealth 0xd38fac5b70c0 3 - 1 False
2025-10-02 20:14:07.000000 UTC N/A Disabled
3720 788 SecurityHealth 0xd38fac5a7080 13 - 0 False
2025-10-02 20:14:07.000000 UTC N/A Disabled
4092 788 WmiApSrv.exe 0xd38fabb94080 3 - 0 False
2025-10-02 20:14:09.000000 UTC N/A Disabled
8132 788 svchost.exe 0xd38fa9c8c080 9 - 0 False
2025-10-02 20:14:12.000000 UTC N/A Disabled
656 7288 chrome.exe 0xd38faae230c0 19 - 1 False
2025-10-02 20:14:13.000000 UTC N/A Disabled
6200 7288 chrome.exe 0xd38fabb7d080 14 - 1 False
2025-10-02 20:14:19.000000 UTC N/A Disabled
7248 7288 chrome.exe 0xd38fabf03080 13 - 1 False
2025-10-02 20:14:19.000000 UTC N/A Disabled
1528 7288 chrome.exe 0xd38fabe1b080 14 - 1 False
2025-10-02 20:14:22.000000 UTC N/A Disabled
6048 7288 chrome.exe 0xd38faba63080 13 - 1 False
2025-10-02 20:14:23.000000 UTC N/A Disabled
5040 7288 chrome.exe 0xd38faba1a080 10 - 1 False
2025-10-02 20:14:23.000000 UTC N/A Disabled
6512 3408 msedge.exe 0xd38fabb9e080 62 - 1 False
2025-10-02 20:14:59.000000 UTC N/A Disabled
8048 6512 msedge.exe 0xd38fac6370c0 8 - 1 False
2025-10-02 20:15:02.000000 UTC N/A Disabled
5724 6512 msedge.exe 0xd38fac6870c0 18 - 1 False
2025-10-02 20:15:03.000000 UTC N/A Disabled
8008 6512 msedge.exe 0xd38fac68e0c0 22 - 1 False
2025-10-02 20:15:03.000000 UTC N/A Disabled
6312 6512 msedge.exe 0xd38fac7590c0 10 - 1 False
2025-10-02 20:15:04.000000 UTC N/A Disabled
0 0 0xd38fa9c07080 0 - N/A False N/A N/A
Disabled
0 232614009783736 0xd38fad177080 0 - - False
N/A N/A Disabled